

### Question 1

a)

1. Total Resistance

$$\begin{aligned} R_T &= R_1 + R_2 + R_3 \\ &= 20\ \Omega + 30\ \Omega + 50\ \Omega \\ &= 100\ \Omega \end{aligned}$$

2. Total Current

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{125\text{V}}{100\ \Omega} \\ &= 1.25\text{A} \end{aligned}$$

3. Current through each resistor

$$\begin{aligned} I_T &= I_1 + I_2 + I_3 \\ &= 1.25\text{A} \end{aligned}$$

4. Voltage drop through each resistor

$$\begin{aligned} P_{20\ \Omega} &= 20\text{V} \times 1.25\text{A} = 31.25\text{W} \\ V_{30\ \Omega} &= 30\text{V} \times 1.25\text{A} = 46.875\text{W} \\ V_{50\ \Omega} &= 50\text{V} \times 1.25\text{A} = 78.125\text{W} \end{aligned}$$

5. Power dissipated

$$\begin{aligned} P &= V \times I \\ P_{20\ \Omega} &= 25\text{V} \times 1.25\text{A} = 31.25\text{W} \\ P_{30\ \Omega} &= 37.5\text{V} \times 1.25\text{A} = 46.875\text{W} \\ P_{50\ \Omega} &= 62.5\text{V} \times 1.25\text{A} = 78.125\text{W} \end{aligned}$$

## Question 2

b

1. Total resistance

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= \frac{8}{100} \quad \left(\frac{1}{R_T}\right)^{-1} = \frac{100}{8} = 12.5 \Omega$$

$$R_T = 12.5 \Omega$$

2. Total current

$$I_T = \frac{125V}{12.5 \Omega}$$

$$= 10A$$

3. Current through each resistor

$$I_{20\Omega} = 125V / 20\Omega = 6.25A$$

$$I_{100\Omega} = 125V / 100\Omega = 1.25A$$

$$I_{50\Omega} = 125V / 50\Omega = 2.5A$$

4. Voltage drop through each resistor

$$= 125V$$

5. Power dissipated

$$P_{20\Omega} = 125V \times 6.25A = 781.25W$$

$$P_{100\Omega} = 125V \times 1.25A = 156.25W$$

$$P_{50\Omega} = 125V \times 2.5A = 312.5W$$